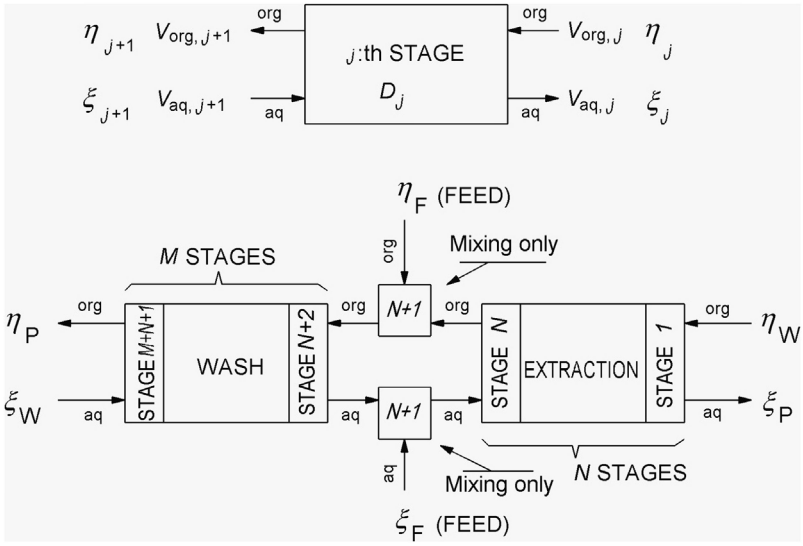


Figure A.4 Notations for counter current extraction with wash stages.

achieved with the counter current, central or intermediate feed solvent extraction technique. This is extensively used in uranium production, nuclear fuel reprocessing, and transuranium element separations. When the conditions are selected so that a high $(P \gg 1)$ extraction factor is obtained for the desired product and low ones $(P \ll 1)$ for the impurities, high purity and good yield can be obtained with relatively few stages. However, θ is always lower in the extraction stages than in the washing stages because the central feed flow rate adds to the wash flow rate in the extraction stages and the organic flow rate is unchanged. Hence, cases with the same P-value in extraction and washing stages seldom occur.

The purpose of the *washing* (or *scrubbing)* stages is to clean the desired product from impurities, and should not be confused with the *stripping* stages, the purpose of which is to transfer the product from the organic phase to a new aqueous phase. However, stripping can also be so arranged that further purification is achieved, sometimes even using a central organic feed, cf. Fig. A.4. If no additional purification is desired after extraction and the extractant is volatile, the organic phase may be distilled, leaving a pure solid product.

A.3. HIGH LOADINGS

A further industrial requirement on solvent extraction processes is high capacity, which means high concentrations of several solutes in the aqueous and organic solvents ('high loading'). The extraction factor P (or distribution factor D_M) may then vary from stage to

stage because D-values and phase volumes change. The calculation of the number of stages, flow rate ratios, etc., needed in order to obtain the desired product has to take this into account.

In order to treat industrial conditions with coupled extraction and washing batteries with central feed where both phase volumes and distribution ratios may change as an effect of the extraction we introduce the mass flow rates: $\xi = X\,V_{\mathrm{aq}}$ and $\eta = Y\,V_{\mathrm{org}}$, where X and Y are the concentrations in the aqueous and organic phases respectively, and V_{aq} and V_{org} are the flow rates of the aqueous and organic phases. We also redefine the extraction factor for the jth stage, P_j, in terms of actual flow rates out of that stage as

$$P_j \;=\; D_j V_{\mathrm{org},j+1} / V_{\mathrm{aq},j} \tag{A.10}$$

where the index $j+1$ means that it flows to the next higher stage and j that it flows to the previous stage (Fig. A.4). Now five useful extraction functions can be defined as follows.

$$e_1 \;=\; \left(1 + \sum_{i=2}^{N} \prod_{k=i}^{N} P_k\right) \tag{A.11}$$

$$e_2 \;=\; \left(1 + \sum_{i=2}^{N} \prod_{k=i}^{N} P_k\right) \tag{A.12}$$

$$s_1 \;=\; \left(1 + \sum_{i=N+2}^{M+N+1} \prod_{k=i}^{M+N+1} P_k\right) \tag{A.13}$$

$$s_2 \;=\; \left(1 + \sum_{i=N+3}^{M+N+1} \prod_{k=i}^{M+N+1} P_k\right) \tag{A.14}$$

$$\zeta \;=\; \prod_{k=N+2}^{M+N+1} P_k \tag{A.15}$$

Through a series of mass balances we can obtain the following equation for mass flow rate with the outgoing aqueous phase, ξ_{P}

$$\xi_{\mathrm{P}} \;=\; [\xi_w + \eta_w(e_2\zeta + s_2) + \xi_{\mathrm{F}} s_1 + \eta_{\mathrm{F}} s_2]/(e_1\zeta + s_2) \tag{A.16}$$

and the outgoing organic phase, η_{P}

$$\eta_{\mathrm{P}} \;=\; \xi_{\mathrm{F}} + \xi_{\mathrm{W}} - \xi_{\mathrm{P}} + \eta_{\mathrm{F}} + \eta_{\mathrm{W}} \tag{A.17}$$

The symbols refer to those given above and in Fig. A.4. Mass flow rates and conditions inside each battery can in principle be obtained recursively from the relations

$$\xi_{n+1} = \xi_n(1 + P_n) - \eta_n \quad \text{and} \quad \eta_{n+1} = \xi_n P_n \tag{A.18}$$

However, numerical problems tend to make the use of these equations difficult for batteries with many stages. Special cases occur when P is either constant or constantly equal 1.

The flow rates out from any stage can usually be calculated with sufficient accuracy from the apparent molar volumes, Φ_i, of the i compounds according to

$$V_{aq} = V_{0,aq} \sum \left(1 + \Phi_{i,aq} X_i\right) \quad \text{and} \quad V_{org} = V_{0,org} \sum \left(1 + \Phi_{i,org}\, Y_i\right) \tag{A.19}$$

where $V_{0,aq}$ and $V_{0,org}$ are the flow rates of the pure unloaded phases in each stage. In many cases one of the central feeds is missing (usually the organic feed) and the corresponding mass and volume flow is zero. The wash stream (index W) is then the real feed for that phase.

When the flow rates vary considerably in the batteries due to loading effects, the whole calculational procedure becomes recursive and a computer code based on these equations may be needed. However, in most cases it is only necessary to treat one, or perhaps two, of the extracted components recursively as it dominates volume changes and defines the P-values for all minor components, e.g. in the high active first stage in the Purex process nitric acid and uranium define the flow rates and extraction factors for all other elements.

In industrial processes it is not only important to extract the desired compound with high yield and purity, but also to be able to strip it again with good efficiency using a minimum amount of chemicals. Contrary to single stage batch extraction, the possibility to vary the P-value by changes in the flow rate ratio (remember that $P = D \times V_{org}/V_{aq}$) makes the use of conditions with D-values for the product not too far from 1 is desirable. In such a case the product can be stripped by a decrease in the flow rate ratio (in more dilute form than in the feed) and a moderate decrease in the D-value without the need to introduce new chemicals for stripping. A typical example is the uranium purification part of the Purex process. Here uranium (and nitric acid) is extracted at high concentration in the presence of 1−6 M HNO_3 and stripped at a low flow rate ratio by more dilute nitric acid. If needed, the separation can be repeated after reconcentration by evaporation.

A.4. SOLVENT EXTRACTION EQUIPMENT

The less conventional part − from a chemical engineering viewpoint − of a reprocessing plant is the solvent extraction equipment, even though the technique is becoming